CSCI 271: Assignment 2

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10 multiple-choice questions covering variables, data types, arithmetic, and input/output from Lesson 2. Circle or check the box next to your answer.

1. What does declaring a variable actually do?

- ☐ A) It immediately sets the value to 0
- ☐ B) It reserves a named location in memory with a specific type
- ☐ C) It prints the variable's name to the screen
- ☐ D) It deletes any variable with the same name

2. Which data type should you use to store a single letter grade, like 'A'?

- ☐ A) int
- ☐ B) string
- ☐ C) char
- ☐ D) double

3. What is the value of int result = 9 / 2;?

- ☐ A) 4.5
- ☐ B) 4
- ☐ C) 5
- ☐ D) It causes a compiler error

4. What does cout << 2 + 3 * 4; print?

- ☐ A) 20
- ☐ B) 9
- ☐ C) 14
- ☐ D) 24

5. What's the difference between << and >>?

- ☐ A) << reads input, >> sends output
- ☐ B) << sends output, >> reads input
- ☐ C) They do the same thing and can be used interchangeably
- ☐ D) << is only for numbers, >> is only for text

6. What does the escape sequence \t do inside a string?

- ☐ A) Starts a new line
- ☐ B) Prints a literal backslash
- ☐ C) Inserts a horizontal tab
- ☐ D) Deletes the previous character

7. Given int x = 5; followed by x = 8;, what happens to the original value 5?

- ☐ A) x now stores both 5 and 8
- ☐ B) x now stores only 8; the 5 is gone
- ☐ C) The program produces a compiler error
- ☐ D) x still stores 5 until the program ends

8. What happens when you write int x = 3.5;?

- ☐ A) It rounds to 4
- ☐ B) It causes a compiler error
- ☐ C) It stores 3, silently dropping the .5
- ☐ D) It stores 3.5 exactly, same as a double would

9. In the line int total = 5;, which word is an identifier rather than a keyword?

- ☐ A) int
- ☐ B) total
- ☐ C) double
- ☐ D) char

10. Why is x = x + 1; valid in C++, even though it looks wrong in math class?

- ☐ A) It's a math error — x can never equal x + 1
- ☐ B) It updates x by taking its current value, adding 1, and storing the result back in x
- ☐ C) It compares x to x + 1 and returns true or false
- ☐ D) It only works if x was never given a value before

Mini Lab: Try It Yourself

// Your task:
// 1. Ask the user to enter their name and age
// 2. Print a greeting using both values
// Example output:
// Hello, I am Choco! I am 1 year old.
Write the C++ program (explain all the lines) and add the screenshot of your output:

```
#include <iostream>
using namespace std;
int main() {
    string name;
    int age;
    cout << "Enter your name: ";
    cin >> name;
    cout << "Enter your age: “;
 cin >> age;
    cout << "Hello, I am " << name << "! I am " << age << " year old." << endl;
    return 0;
}
```

Line Explanation:
#include <iostream>
This lets us use cout and cin in our program.
using namespace std;
This lets us use C++ commands like cout, cin, and string more easily.
int main() {
This is where the program starts.
string name;
This creates a variable called `name` to store the user's name.
int age;
This creates a variable called age to store the user's age.
cout << "Enter your name: ";
This asks the user to enter their name.
cin >> name;
This takes the name the user types and saves it in name
cout << "Enter your age: ";
This asks the user to enter their age.
cin >> age;
This takes the age the user types and saves it in age
cout << "Hello, I am " << name << "! I am " << age << " year old." << endl;
This prints the greeting using the name and age the user entered.
return 0;
This tells the computer that the program finished successfully.
}
This shows that the program is finished.